DAY 06 OF 16 · 3D & 6DOF STREAMING

How 3D is captured

LiDAR, photogrammetry and depth cameras, and the one formula that links a pixel to a point in space.

Sensors | Pinhole camera | Intrinsics & extrinsics | RGB-D → points | Fusing views

Book: Ch. 1, 3, 10 · Code: day06_capture_camera_model/

BOOK CH. 1 & 3

Three ways to capture the real world in 3D

LiDAR

Measures distance with laser pulses. Accurate, works in the dark; terrestrial, mobile and airborne.

01

Photogrammetry

Many overlapping photos → Structure from Motion → dense points and meshes. Cheap; struggles with glass and shiny metal.

02

Depth cameras

Structured light or time of flight give RGB-D frames in real time (Kinect, RealSense, phone LiDAR).

03

THE PINHOLE CAMERA

From a 3D point to a pixel

$$u = f_x \cdot X / Z + c_x \quad v = f_y \cdot Y / Z + c_y$$

X, Y, Z

Point in CAMERA coordinates (after applying the pose).

fx, fy

Focal length in pixels: how much the image is zoomed.

cx, cy

Principal point: usually the image centre.

BOOK CH. 10 NOTE

Intrinsics vs extrinsics

Intrinsics K

Inside the camera: focal length, principal point, lens distortion. Found by calibration with a checkerboard.

01

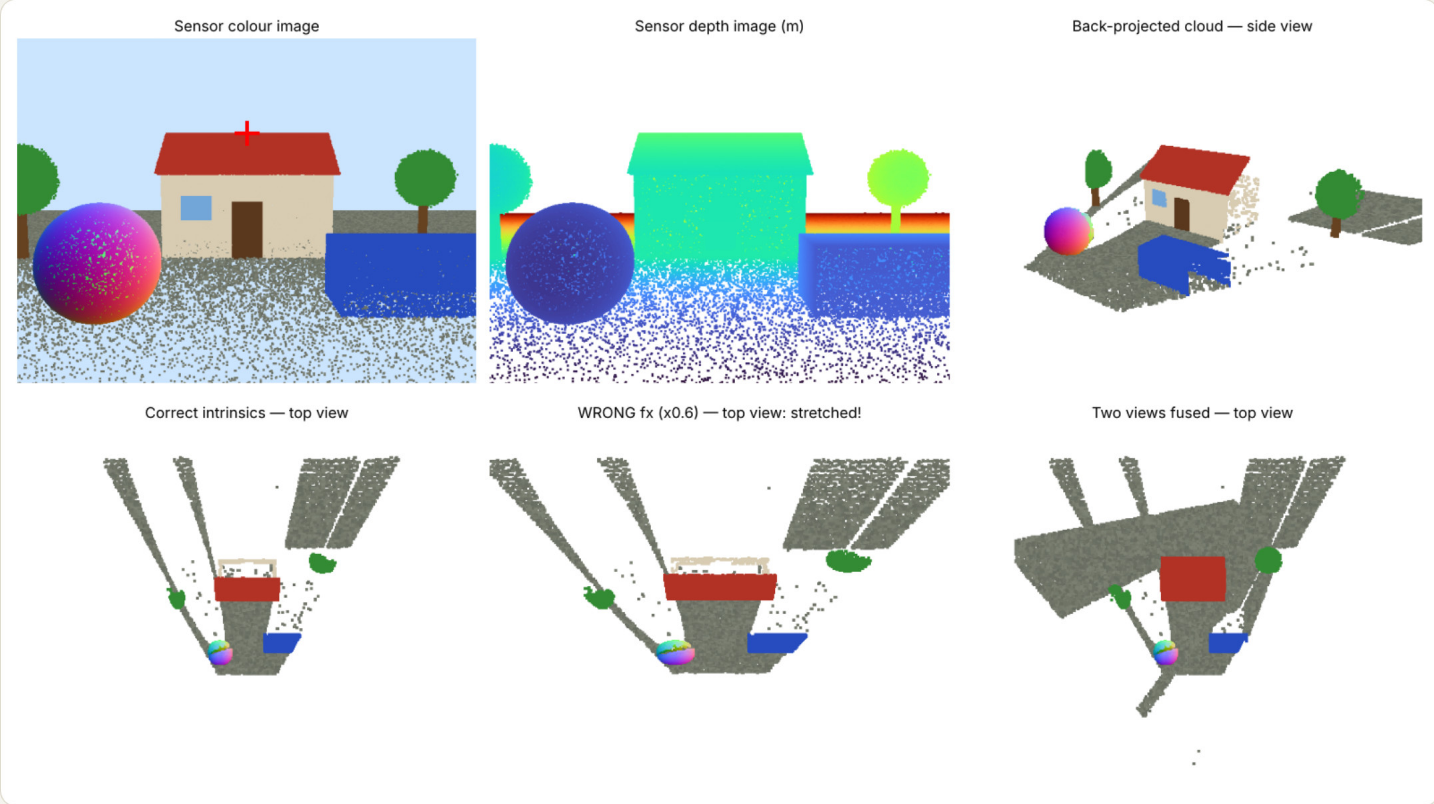
Extrinsics [R | t]

Where the camera is and where it looks: the 6DoF pose from Day 2.

02

LAB PREVIEW

Depth image → point cloud, and why K must be right



With the right fx the house is 6.5 m wide; with fx × 0.6 it becomes 10.8 m. Two views fill the hidden side.

HANDS-ON LAB · COPY, RUN, LOOK

Build a point cloud from a (simulated) depth camera

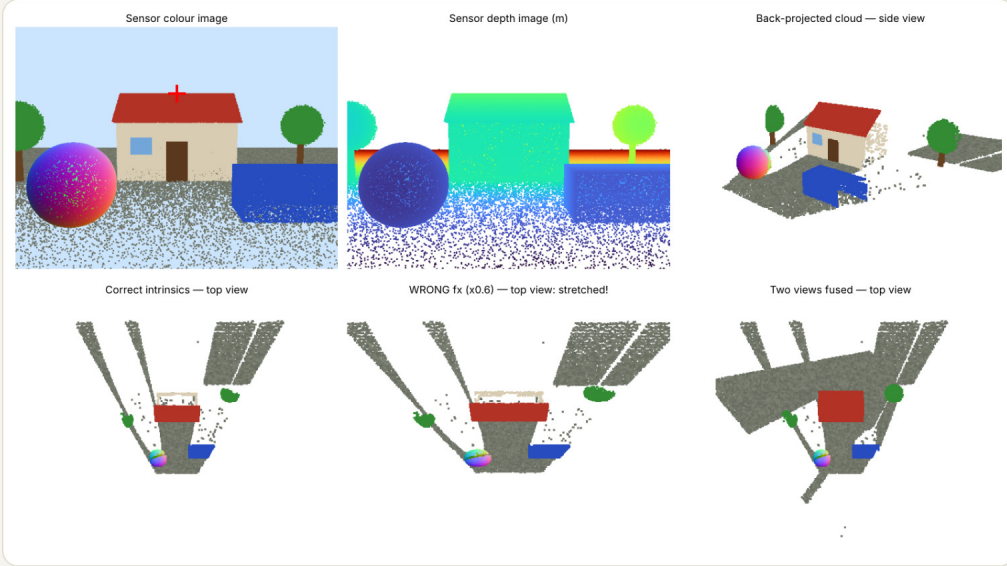
RUN IT

```
cd day06_capture_camera_model
python pinhole_and_rgbd.py
```

→ Check the hand-computed pixel (u, v).

→ Compare house widths: right vs wrong fx.

→ Open outputs/fused_two_views.ply.



Three things to remember

1. LiDAR, photogrammetry and depth cameras all end as points with positions.
2. Pixel = K · [R | t] · point. Intrinsics inside, extrinsics = the 6DoF pose.
3. One view is 2.5D; fusing several views gives full 3D.

BEFORE NEXT SESSION

Lab L3 and mini-project P1 (scan to cloud). Next: organising millions of points.